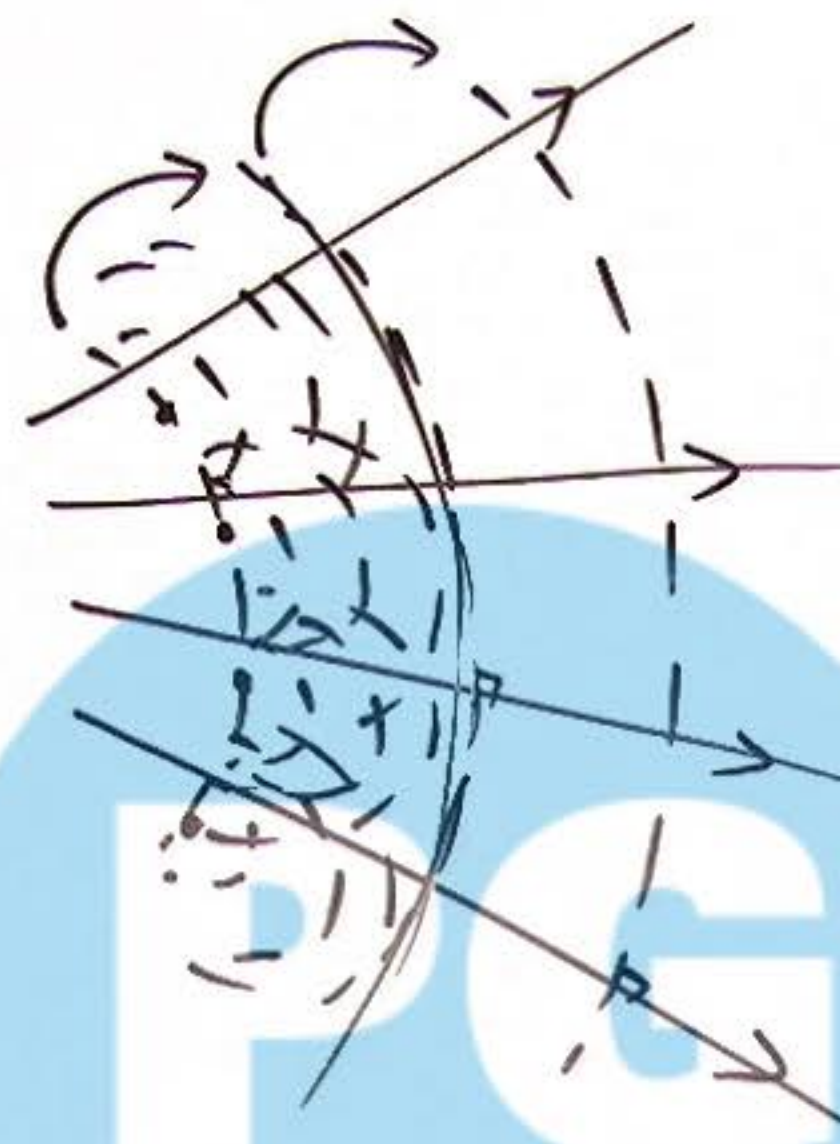
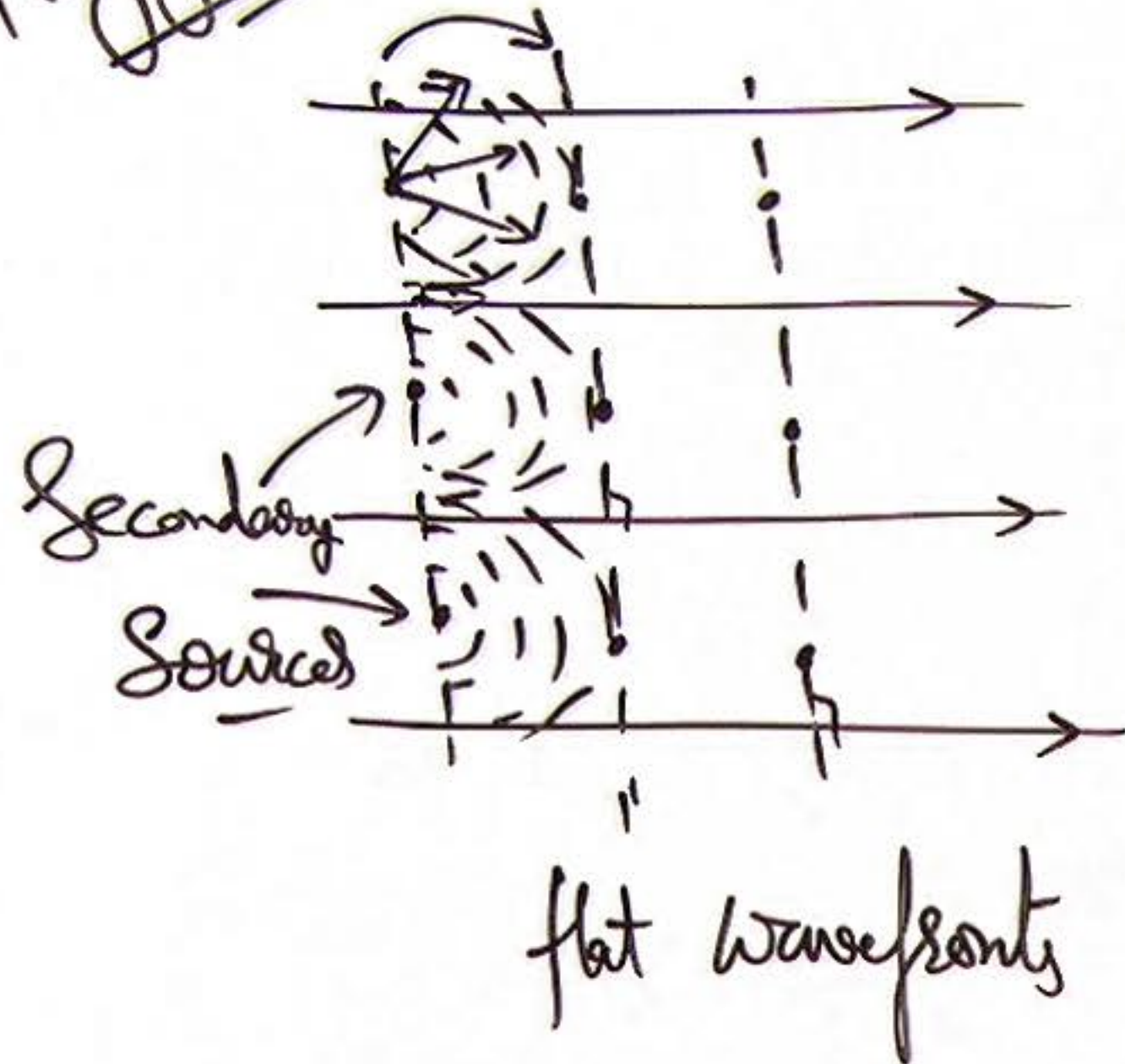
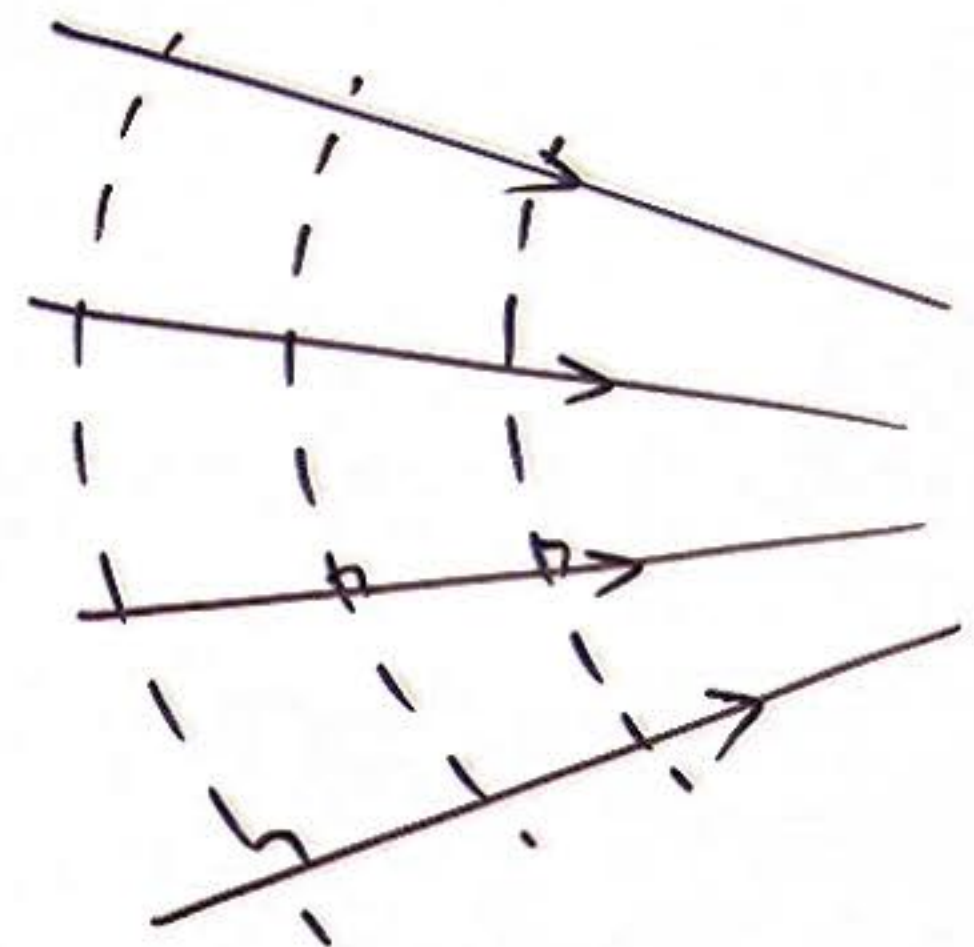


① Wavefront of a Light Wave :

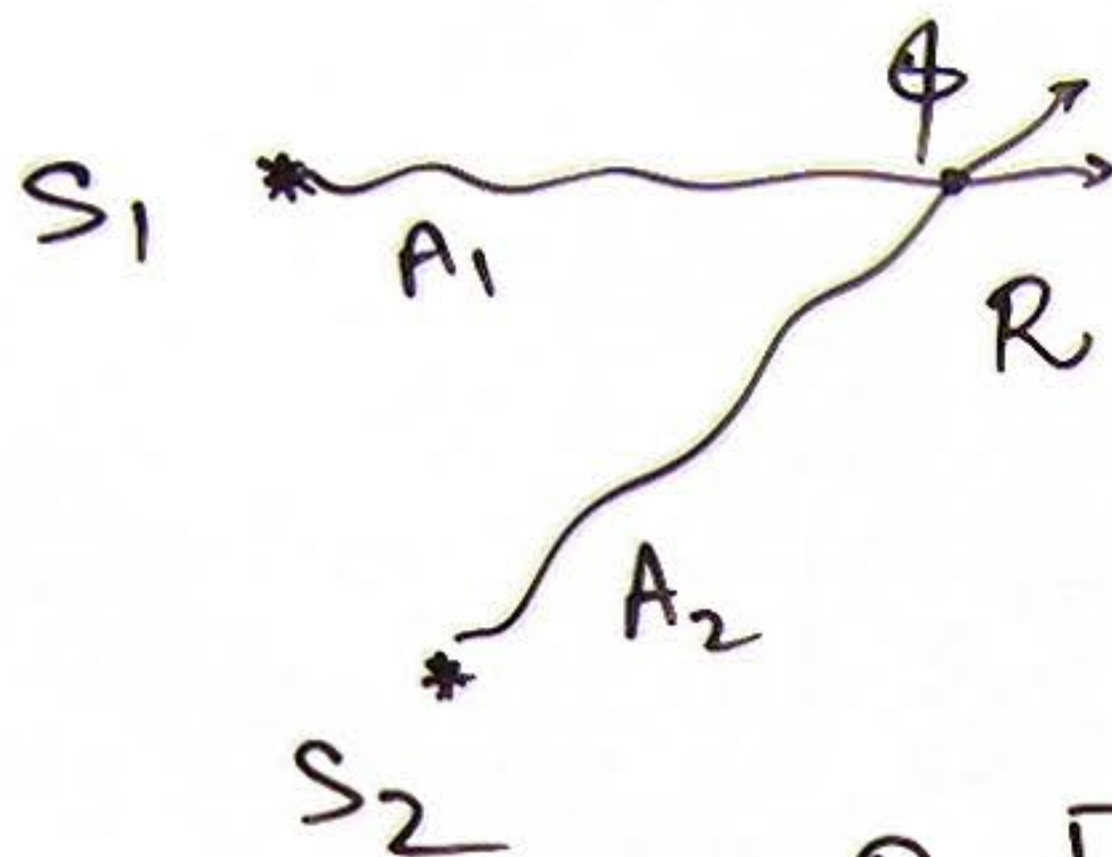
Huygens's theory of wave propagation



Conver W/F
Physics Galaxy
Videos



Concave W/F

② Interference of Light:Condition $\rightarrow$ coherence.

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$\text{if } A_1 = A_2 = A \Rightarrow R = 2A \cos \phi / 2$$

③ Intensity at pt of Interference

$$I = kR^2$$

$$I_R = kA_1^2 + kA_2^2 + 2kA_1A_2 \cos \phi$$

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\text{if } I_1 = I_2 = I_0$$

$$I_R = 4I_0 \cos^2 \phi / 2$$

④ Constructive & Destructive Interference:Construct IntCond $\rightarrow$ Same phase

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\phi = 2N\pi$$

$$R = A_1 + A_2 = 2A$$

$$I_R = (\sqrt{I_1} + \sqrt{I_2})^2 = 4I_0$$

path diff

$$\Delta = \frac{\lambda}{2\pi} \cdot \phi = N\lambda$$

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VideosDestruct IntCond $\rightarrow$ opp phase

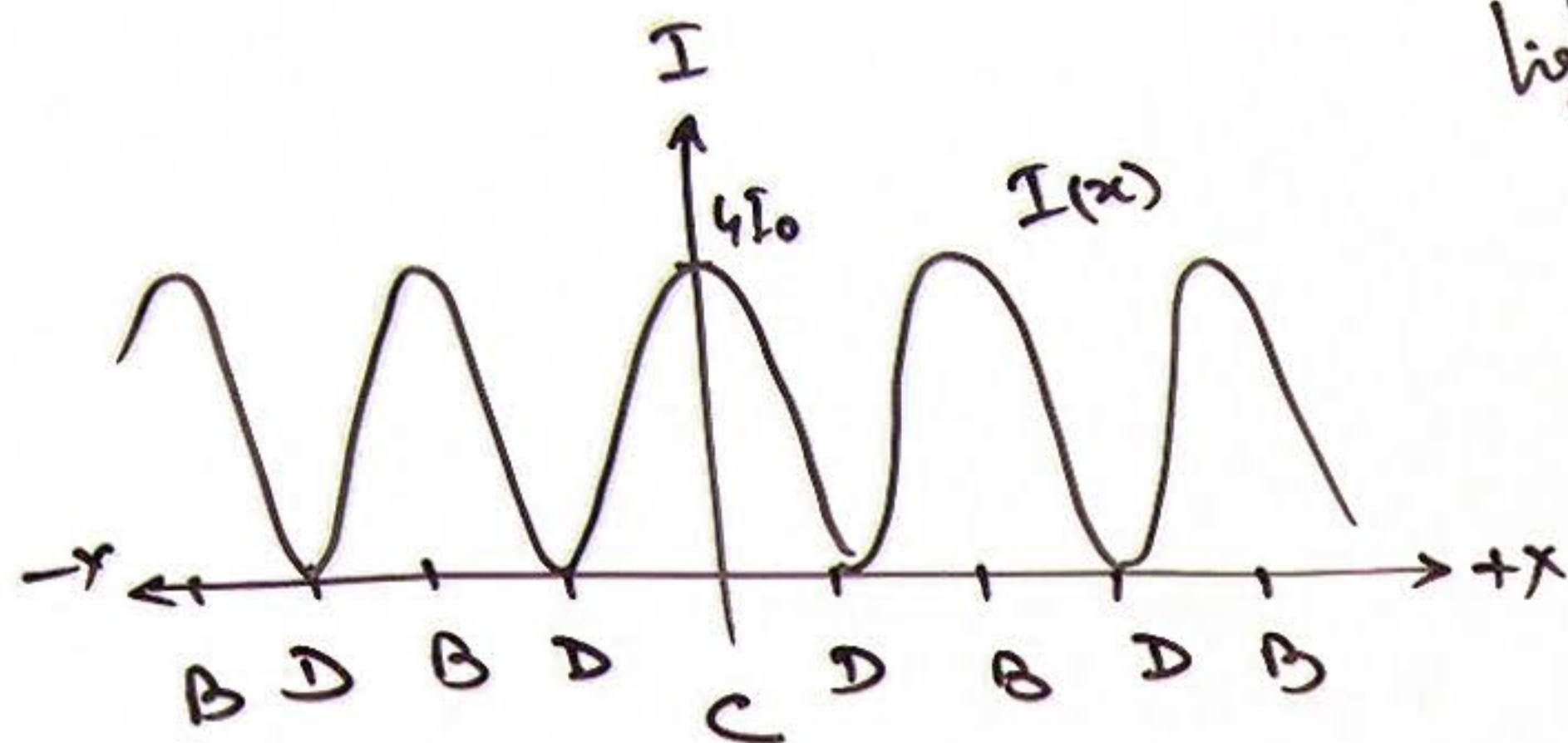
$$\phi = (2N+1)\pi$$

$$R = |A_1 - A_2| = 0$$

$$I_R = (\sqrt{I_1} - \sqrt{I_2})^2 = 0$$

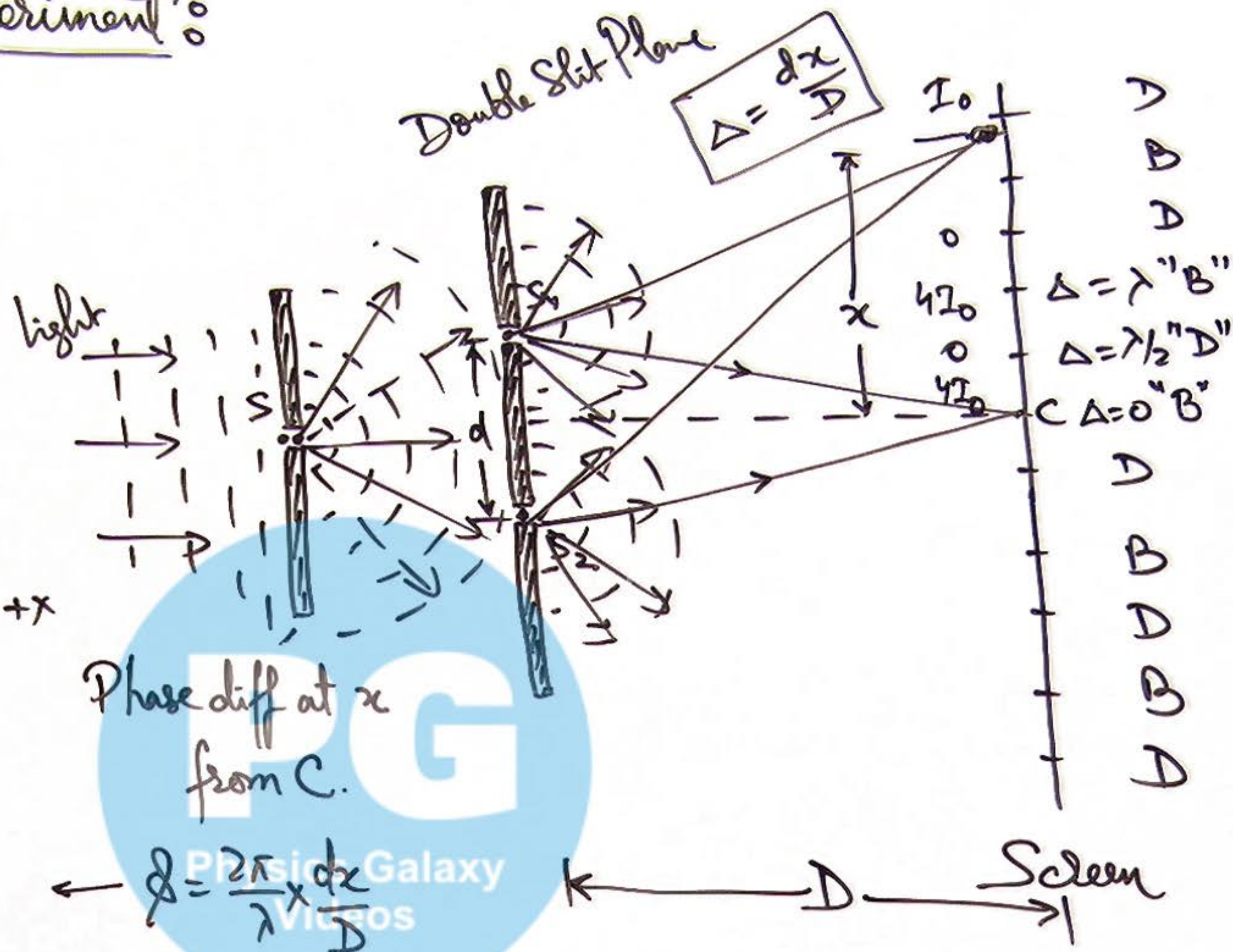
$$\Delta = (2N+1)\frac{\lambda}{2}$$

⑤ Young's Double Slit Experiment:



$$I(x) = 4I_0 \cos^2 \frac{\phi}{2}$$

$$I(x) = 4I_0 \cos^2 \left(\frac{\pi d x}{\lambda D} \right)$$



⑥ Position of Bright & Dark fringes in YDSE:

distance of n^{th} bright fringe from C

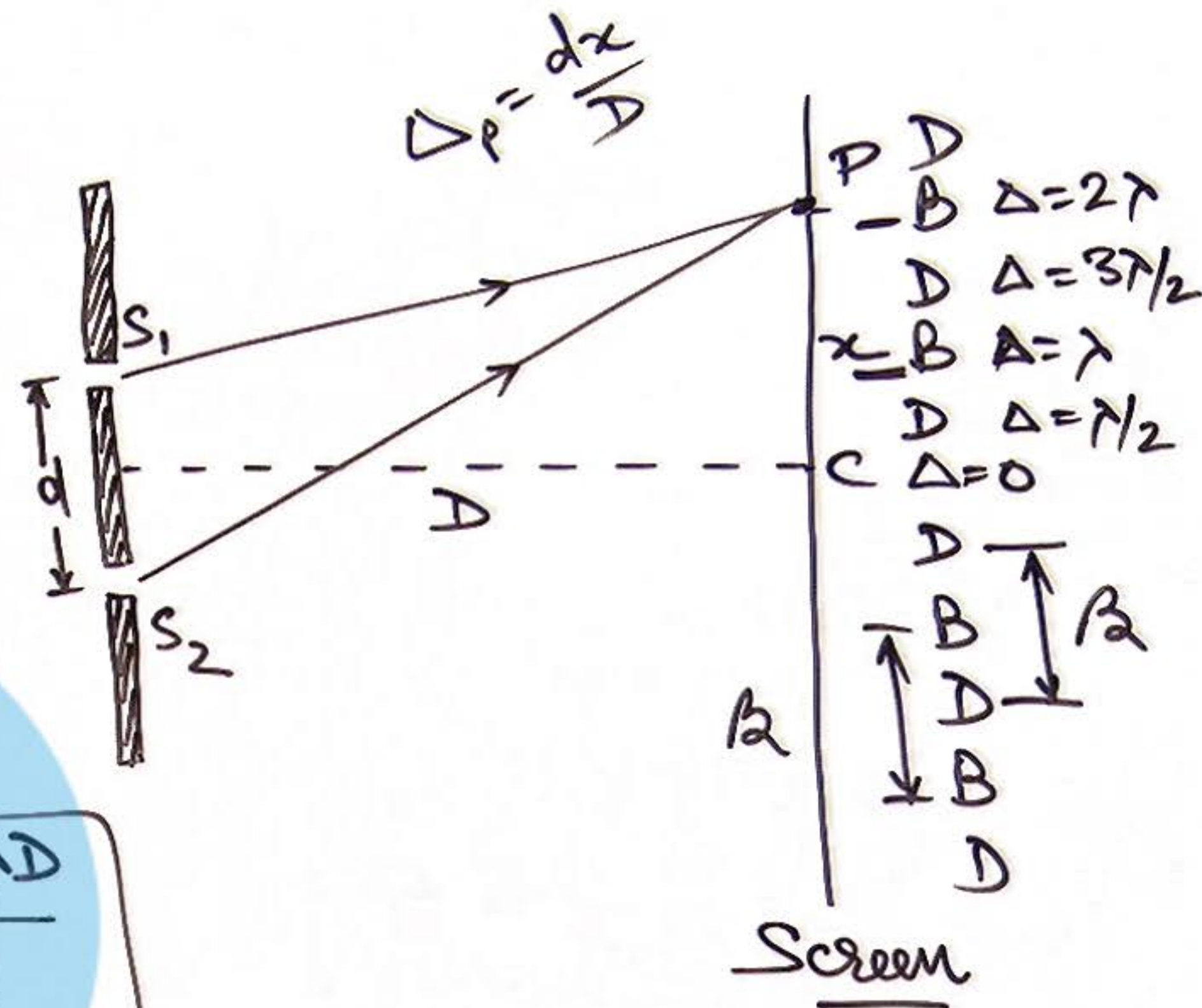
$$\frac{dx_{nB}}{D} = n\lambda \Rightarrow \boxed{x_{nB} = \frac{n\lambda D}{d}}$$

for dark fringe

$$\frac{dx_{nD}}{D} = (2N-1)\frac{\lambda}{2} \Rightarrow \boxed{x_{nD} = \frac{(2N-1)\lambda D}{2d}}$$

⑦ Fringe Width:

$$\boxed{\beta = x_{nB} - x_{(n-1)B} = \frac{\lambda D}{d}}$$

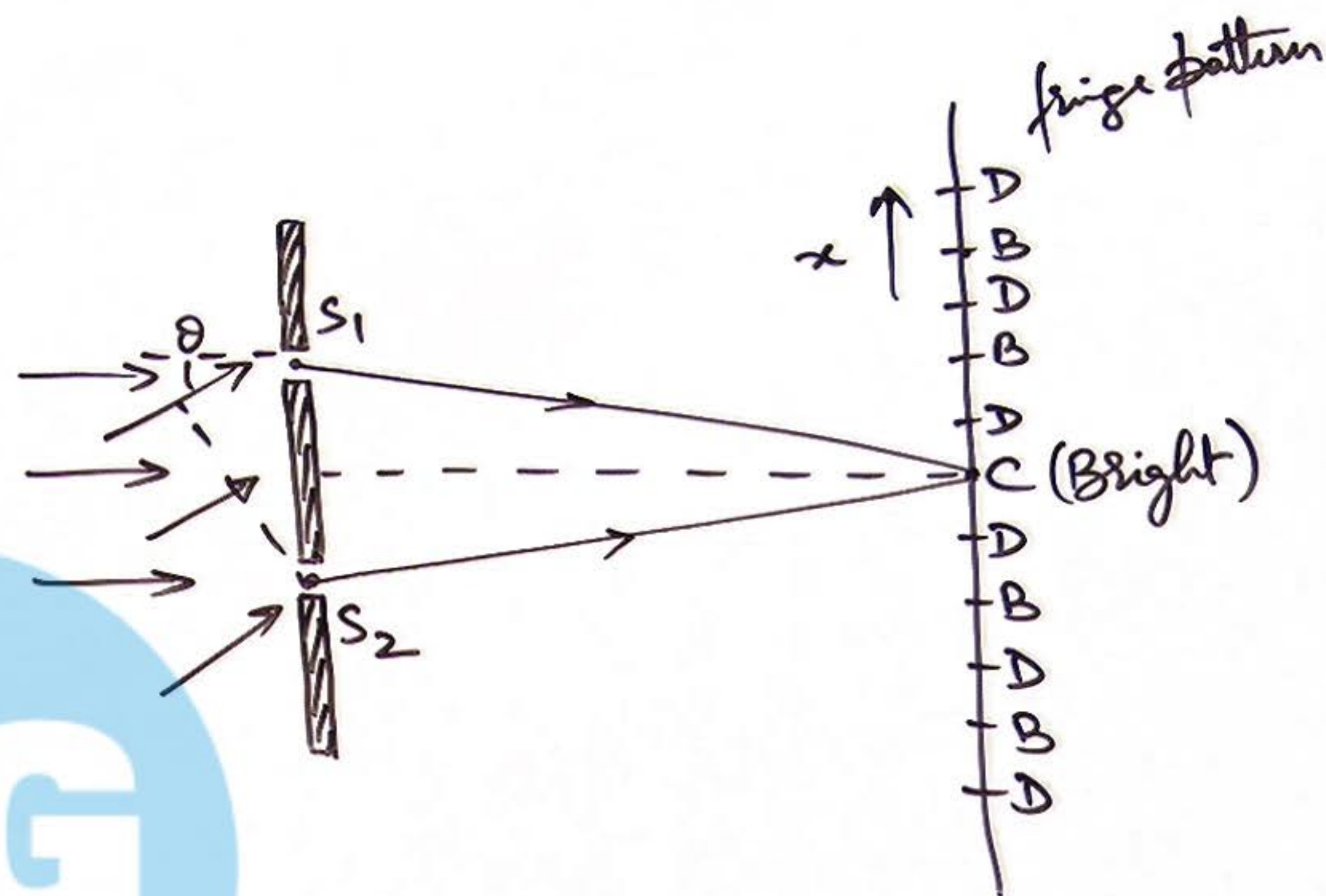


⑧ Changing Direction of Incident
Light in YDSE:

$$\frac{dx}{D} = d \sin \theta$$

$$x = D \sin \theta$$

↑
Shift of fringe pattern



PG

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⑨ Effect of Submerging YDSE Setup
in a Transparent Medium :

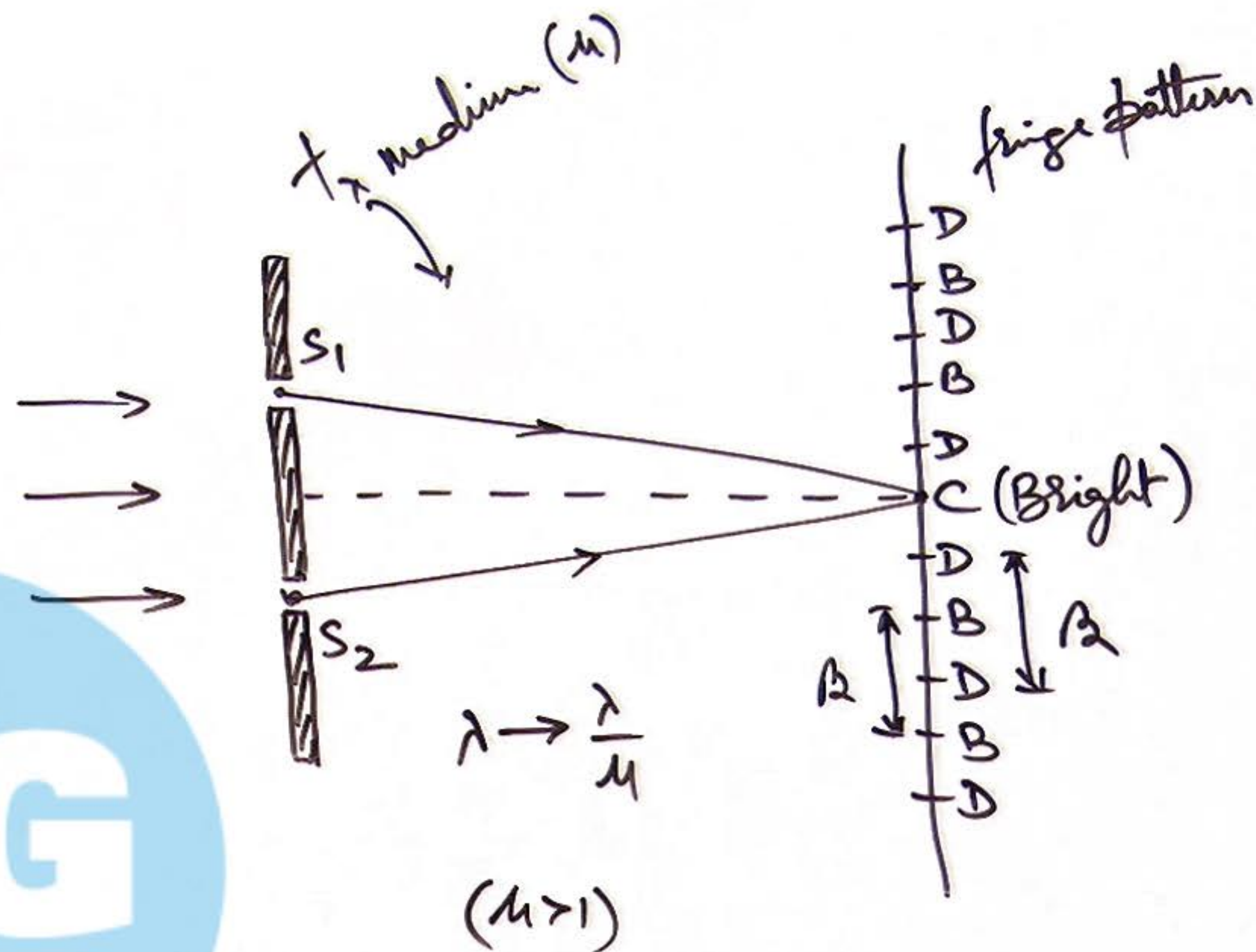
fringe width $\beta = \frac{\lambda D}{d}$

$$\beta' = \frac{\lambda D}{\mu d} = \left(\frac{\beta}{\mu} \right)$$

$\beta' < \beta$

PG

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⑩ Effect of insertion of a thin film in YDSE:

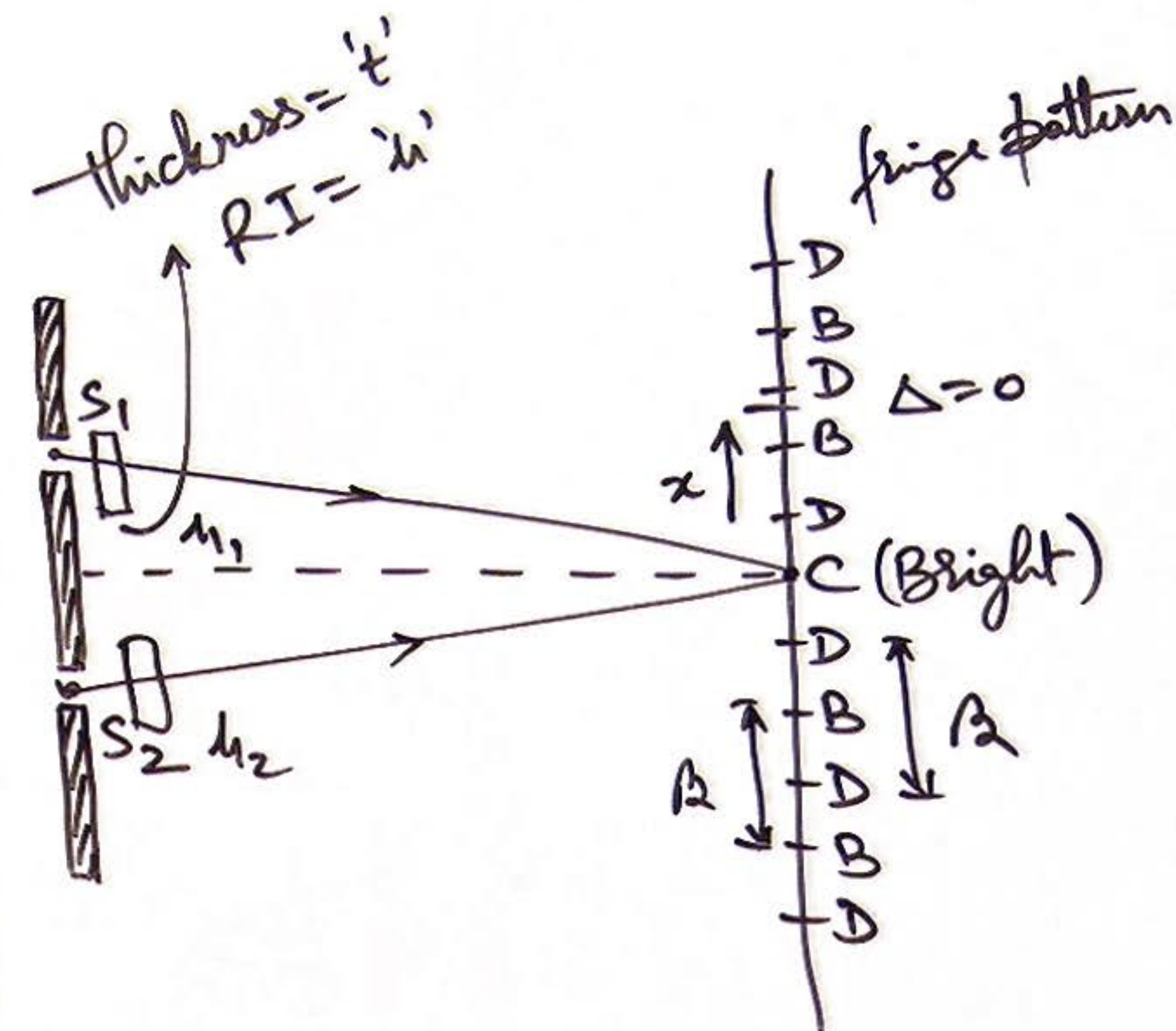
✓ Shift of fringe pattern

φ.d due to film φ.d. at x

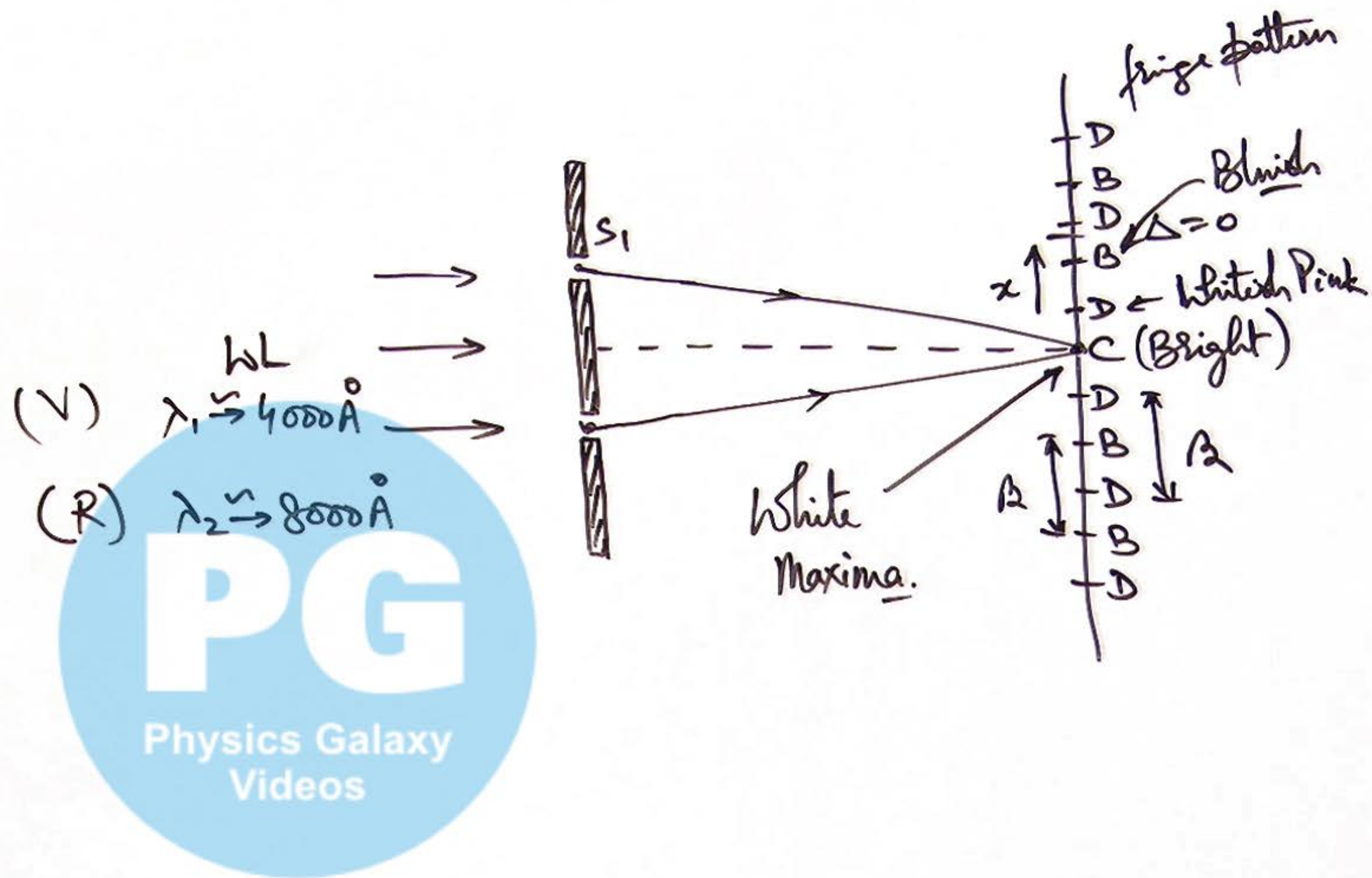
$$t(\mu-1) = \frac{dx}{D}$$

$$\Rightarrow x = \frac{Dt(\mu-1)}{d}$$

PG
Physics Galaxy
Videos



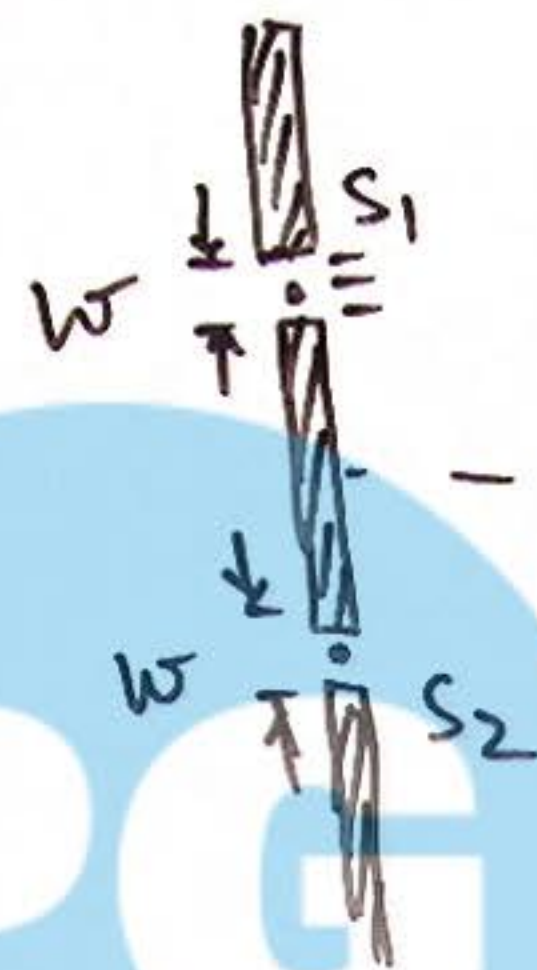
⑪ Use of White Light in YDSE :



⑫ Effect of Changing Slit width in YDSE:

Contrast ratio $\mathcal{R}_c = \left[\frac{I_{\max}}{I_{\min}} \right]$

$$\mathcal{R}_c = \left[\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right]^2$$



$$\begin{aligned} \sqrt{I_1} &\propto w_1 \\ \sqrt{I_2} &\propto w_2 \end{aligned}$$

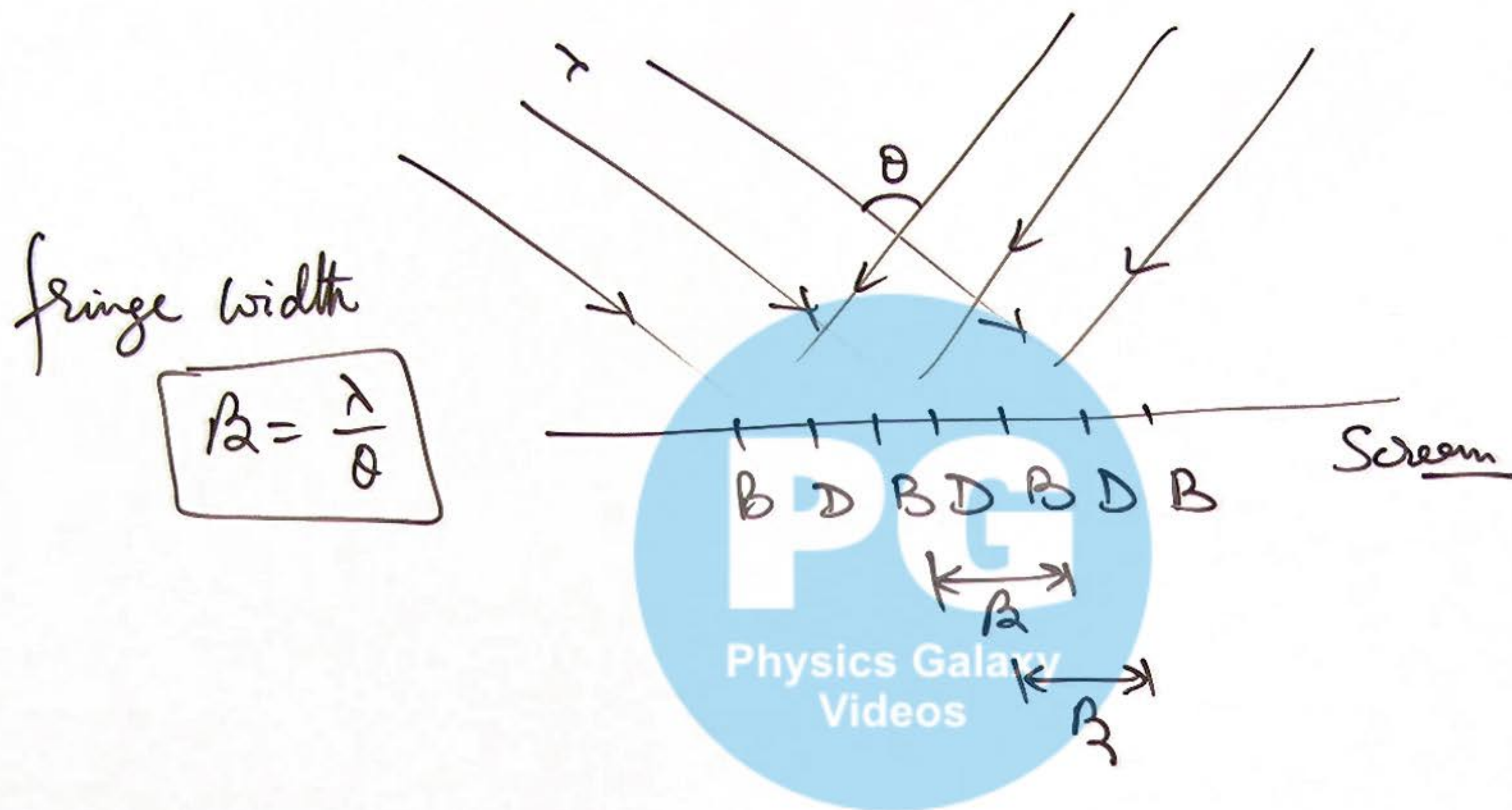
$$I_D = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$D(0)$$

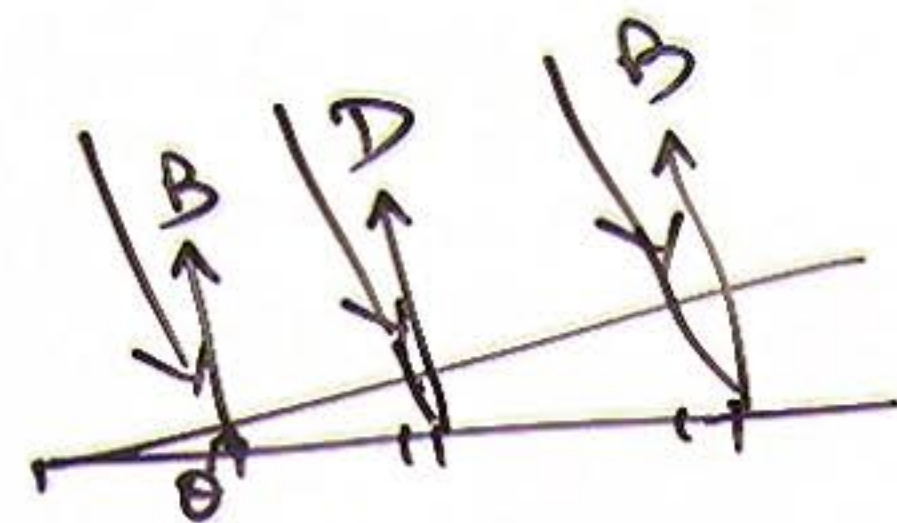
$$B(4I_0)$$

$$I_B = (\sqrt{I_1} + \sqrt{I_2})^2$$

(13) Interference of two Converging
Coherent parallel beams:



①④ Interference by thin films:



①⑤ Interference due to a wedge

①⑥ Interference due to a lens.

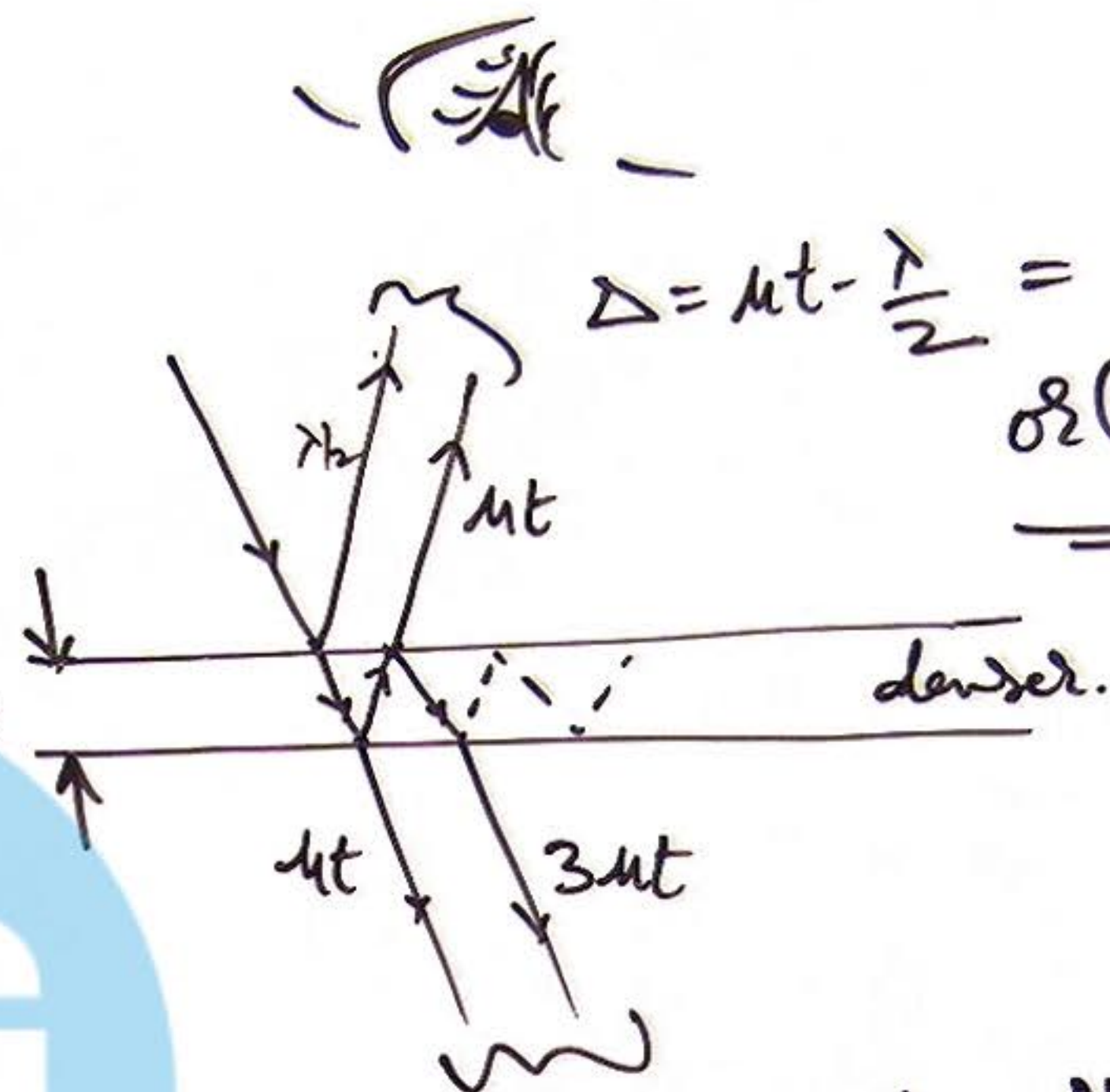


(Newton's Rings)

Plano Convex lens

①⑦ Shape of Interference fringes.

μ, t



$$\Delta = \mu t - \frac{\lambda}{2} = N\lambda$$

$$\text{or } (2N+1)\frac{\lambda}{2}$$

$$\Delta = 2\mu t = N\lambda$$

$$\text{or } (2N+1)\frac{\lambda}{2}$$